

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 11- Integration

September, 2023.

1. Find the following integrals:

(a) $\int (2x + 3)x^2 \, dx$	(c) $\int \left(x + \frac{1}{x}\right)^2 \, dx$	(e) $\int \left(3 + \frac{\sqrt{x+6x^3}}{x}\right) \, dx$
(b) $\int \sqrt{x}(x + 2) \, dx$	(d) $\int (\sqrt{x} + 2)^2 \, dx$	(f) $\int \frac{(2x+1)^2}{\sqrt{x}} \, dx$

2. The gradient of a curve is given by $f'(x) = x^2 - 3x - \frac{2}{x^2}$. Given that the curve passes through the point $(1, 1)$, find the equation of the curve in the form $y = f(x)$.

3. Given that $y^{\frac{1}{2}} = x^{\frac{1}{3}} + 3$:

- (a) Show that $y = x^{\frac{2}{3}} + Ax^{\frac{1}{3}} + B$, where A and B are constants to be found.
(b) Hence find

$$\int y \, dx$$

4. Evaluate each of the following:

(a) $\int_1^8 (x^{-\frac{1}{3}} + 2x - 1) \, dx$	(c) $\int_1^4 (\sqrt{x} - 3)^2 \, dx$	(e) $\int_0^1 x^2 \left(\sqrt{x} + \frac{1}{x}\right) \, dx$
(b) $\int_1^3 \left(\frac{x^3+2x^2}{x}\right) \, dx$	(d) $\int_3^6 \left(x - \frac{3}{x}\right)^2 \, dx$	(f) $\int_1^4 (x^{\frac{1}{2}} - 4)(x^{-\frac{1}{2}} - 1) \, dx$

5. Find the area between the curve with equation $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$ in each of the following:

- (a) $f(x) := x(x + 2)(x - 2)$; $a = 0, b = 2$.
(b) $f(x) := 2 + 2x$; $a = 1, b = 4$.
(c) $f(x) := x^3$; $a = -4, b = 4$.

6. Sketch the graph of $y = x(x^2 - 4)$, and hence find the area of the region, above the x -axis, bounded by the curve and the x -axis.

7. Find the area of the finite region between the curve with equation $y = 1 + x$, $x = -3$, $x = 3$ and the x -axis.

8. Find the following integrals using substitution method:

$$\begin{array}{lll}
\text{(a)} \int \frac{x+1}{\sqrt{2x^2+4x}} dx & \text{(d)} \int 2xe^{x^2-3} dx & \text{(g)} \int_0^5 x^3\sqrt{x^4+1} dx \\
\text{(b)} \int 2x \sin(x^2+1) dx & \text{(e)} \int \frac{\cos x}{x+\sin x} dx & \text{(h)} \int_0^{\frac{\pi}{2}} \cos x \sin x dx \\
\text{(c)} \int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx & \text{(f)} \int e^{\sin x} \cos x dx & \text{(i)} \int_{-1}^1 (3x-1)\sqrt{3x^2-2x+3} dx
\end{array}$$

9. Find the following integrals using substitution method:

$$\begin{array}{lll}
\text{(a)} \int \frac{x+1}{\sqrt{2x^2+4x}} dx & \text{(d)} \int 2xe^{x^2-3} dx & \text{(g)} \int_0^5 x^3\sqrt{x^4+1} dX \\
\text{(b)} \int 2x \sin(x^2+1) dx & \text{(e)} \int \frac{\cos x}{x+\sin x} dx & \text{(h)} \int_0^{\frac{\pi}{2}} \cos x \sin x dx \\
\text{(c)} \int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx & \text{(f)} \int e^{\sin x} \cos x dX & \text{(i)} \int_{-1}^1 (3x-1)\sqrt{3x^2-2x+3} dx
\end{array}$$

10. Show that

$$\begin{array}{ll}
\text{(a)} \int_1^2 \frac{x^7-2x+1}{4x^3} dx = \frac{223}{160} & \text{(c)} \int_0^{\frac{\pi}{2}} \cos x \sin^5 x dx = \frac{1}{6} \\
\text{(b)} \int_0^1 \frac{x}{(2x^2+1)^3} dx = \frac{1}{9} & \text{(d)} \int_0^{\frac{\pi}{4}} \tan x \sec^2 x dx = \frac{1}{2}
\end{array}$$

11. Using integration by parts, integrate

$$\begin{array}{lll}
\text{(a)} \int x^2 \cos x dx & \text{(d)} \int \frac{\ln x}{x^2} dx & \text{(g)} \int xe^{2x} dx \\
\text{(b)} \int x^3 \sin x dx & \text{(e)} \int x^2 \ln x dx & \text{(h)} \int x^2 e^{4x} dx \\
\text{(c)} \int \ln x dx & \text{(f)} \int (x+1)^2 \ln 3x dx & \text{(i)} \int e^{2x} \cos x dx
\end{array}$$

12. Use the method of partial fraction to integrate the following:

$$\begin{array}{lll}
\text{(a)} \int \frac{x}{(2-x)(3+x)} dx & \text{(c)} \int \frac{x^2+x+5}{x^3+2x^2+x} dx & \text{(e)} \int_1^2 \frac{1}{x(3x-1)^2} dx \\
\text{(b)} \int \frac{6x+13}{x^2+5x+6} dx & \text{(d)} \int_1^2 \frac{3}{x(x+1)} dx & \text{(f)} \int_0^2 \frac{1}{(x+1)(x+2)} dx
\end{array}$$

13. Given that the marginal revenue MR is the first derivative of the revenue function R . Find the revenue function R given that $MR = 100 + 2x - 5x^2$ and $R = 0$, when $x = 0$.

14. The marginal cost function of a product is given by $dC/dx = 100 + 10x + 0.1x^2$ where x is the output. Obtain the total and the average cost function of the firm under the assumption, that its fixed cost is ZMK500.